

Sell Side Equity Report

Teradyne Inc.
(NASDAQ: TER)

Price: **\$74.21 (May 5, 2025)**

Rating: **Buy**

Target Price: **\$89.79**



Earnings Call

Most recent earnings call: April 29, 2025

Next Earnings call: July 28, 2025

Financial Metrics

Sector	Information Technology
Industry	Semiconductor Equipment
Market Capitalisation	\$13,059.8M
52 Week Range	\$65.77 - \$163.205
ROIC	20.17
Price /Earnings (TTM)	24.6
Beta	1.58
Price / Book	4.68
Price / Sales	4.5
Dividend Yield	0.6%
Projected Upside	21%
1 Year Return	-21.20%
3 Year Return	19.22%

DuPont Analysis

Tax Burden	89.05
Interest Burden	100.60
Operating Margin	21.47
Asset Turnover	0.78
Leverage Ratio	1.35
Adjusted ROE	13.05

Price Movement



Company Overview

Teradyne, Inc. is a leading global supplier of automated test equipment and robotics solutions. The company designs, develops, manufactures, and sells automated test systems and robotics products. Teradyne's automated test systems are used to evaluate semiconductors, wireless products, data storage devices, and complex electronic systems across various industries, including consumer electronics, wireless, automotive, industrial, computing, communications, and aerospace and defense. Its robotics portfolio includes collaborative robotic arms and autonomous mobile robots, which are utilized by global manufacturing, logistics, and industrial customers to enhance quality, improve manufacturing and material handling efficiency, and reduce manufacturing and logistics costs. Teradyne's offerings in automated test equipment and robotics products and services comprise:

1. Semiconductor test
2. Robotics
3. Defense/Aerospace

Industry Overview

The market for test products is concentrated with a limited number of significant customers accounting for a substantial portion of the purchases of test equipment. A few customers drive significant demand for the products both through direct sales and sales to the customer's supply partners. The semiconductor testing industry is essential for ensuring chip quality and performance across consumer, automotive, and industrial applications. Valued at around \$8.5 billion in 2023, it is expected to grow at a 5–7% CAGR, driven by demand from AI, 5G, EVs, and advanced packaging. The industry is cyclical and capital-intensive, dominated by players like Teradyne, Advantest, and Cohu. Increasing chip complexity and shrinking nodes are accelerating the need for system-level testing and AI-driven automation. As semiconductor supply chains diversify globally, testing companies are positioned for long-term growth.

Teradyne: A Mission-Critical Enabler in Semiconductors and Automation

Teradyne, Inc. is a mission-critical enabler of the semiconductor and industrial automation revolutions. The company is uniquely positioned to capitalize on long-term structural growth trends in artificial intelligence (AI), electric vehicles (EVs), datacentres, and robotics.

1. Semiconductors: The Bottleneck of Innovation

Modern AI/ML chips contain tens of billions of transistors, pushing complexity to new limits at cutting-edge nodes. The need for high-precision, high-performance testing has never been greater, as defects are harder to detect and far costlier if missed. Teradyne's UltraFLEX and Eagle platforms are purpose-built to meet these demands through industry-leading automated test equipment (ATE). In mission-critical sectors—EVs, ADAS, and medical devices—chip failure can have life-threatening consequences. Devices must withstand extreme conditions like temperature swings, vibration, and radiation. Robust testing isn't optional—it's essential. Advanced chips—AI accelerators, NPUs, and 5G modems—also run at far higher frequencies and thermal loads, often 10x those of legacy designs. Teradyne's thermal-aware platforms ensure these chips meet reliability standards. Semiconductor testing is the gateway to innovation—no chip ships without it. At this critical chokepoint, Teradyne safeguards trust in technology and captures value with every leap forward.

2. Diversified End-Market Exposure

Teradyne's diversified revenue mix enhances its resilience and long-term growth potential. Roughly 70% of revenue comes from semiconductor testing across high-growth segments like mobile, AI/datacenters, and automotive, with major customers including Apple, NVIDIA, Qualcomm, and Intel. As chip complexity rises—especially in AI and automotive—demand for Teradyne's test solutions continues to grow. Its industrial automation segment, contributing nearly 20% of revenue, is expanding through subsidiaries Universal Robots (cobots) and Mobile Industrial Robots (MiR), which saw over 50% YoY growth in 2023. The company is also moving into biotech and medical device automation for precision lab tasks. Geographically, Teradyne benefits from strong exposure to Asia's semiconductor hubs and tailwinds from U.S. reshoring and European automation demand. This broad market reach positions the company to capitalize on multiple secular growth trends while mitigating reliance on any single industry or region.

AHA Moment

Teradyne: The Gatekeeper of Innovation

Every single AI chip, EV power module, or datacentre processor that will define the next decade must first pass through Teradyne's test systems. No test, no tech. Teradyne is the silent gatekeeper of the next industrial revolution. Without Teradyne, the future's most advanced technologies quite literally cannot launch. Teradyne isn't just riding the semiconductor wave—it's the quality gatekeeper for the entire digital revolution. This encapsulates Teradyne's indispensable role: Before the world's most transformative technologies can change our lives, they must first be validated by Teradyne. This "picks-and-shovels" positioning means Teradyne profits from every major tech trend, regardless of which end-market player wins. Teradyne is a high-conviction investment for those seeking exposure to the foundational infrastructure of tomorrow's technology—AI, EVs, robotics, and beyond.

Scaling Innovation—One Acquisition at a Time

With a commanding market share and deep cash flows from this high-margin segment, it's not just riding the innovation wave—it's funding the next one. Years ago, Teradyne saw around the corner and acquired Universal Robots, planting a flag in industrial automation long before cobots became a megatrend. That move wasn't a one-off. It was a blueprint. Now, with the AI, EV, and automation booms accelerating—and a fragmented field of emerging robotics and testing startups—Teradyne is uniquely positioned to repeat that playbook. It can identify, acquire, and scale the next generation of automation pioneers, compounding its lead in test and robotics. The realization? Teradyne isn't just a semiconductor test company. It's a quiet consolidator of automation infrastructure—profitably bridging the old tech stack with the future of smart factories, intelligent machines, and connected devices.

Valuation

Teradyne Inc. presents a compelling upside opportunity as a leading provider of automated test equipment (ATE) and industrial automation solutions. Based on an EV/EBITDA multiple valuation, we arrive at a target price of \$89.79, reflecting ~21% upside from the current share price of \$74.21.

Applying a target 2025E EBITDA of \$950M and a justified 18x EV/EBITDA multiple, we derive:

Enterprise Value (EV): \$17.1B

Equity Value (incl. net cash): \$18.5B

Implied Share Price: \$89.79

Given Teradyne's high free cash flow conversion, zero long-term debt, and strong market positioning in mission-critical testing, EV/EBITDA is the most appropriate valuation method. For FY2024, Teradyne reported:

Revenue: \$2.90B

EBITDA: \$830M (EBITDA Margin: ~28.6%)

Net Cash: ~\$1.4B

EV/EBITDA-Based Sensitivity Analysis

EV/EBITDA	ENTERPRISE VALUE (\$M)	EQUITY VALUE (\$M)	IMPLIED SHARE PRICE (\$)
15	14250	15650	76.34
16	15200	16600	80.98
17	16150	17550	85.61
18	17100	18500	89.79
19	18050	19450	94.88
20	19000	20400	99.51
21	19950	21350	104.15

Peer Comparison:

Company	EV/EBITDA	PE	EV/Sales
Teradyne	18x	24.5x	5.1x
Advantest	39.7x	36.2x	9.4x
AMAT	13.8x	19.8x	3.8x
Keysight	16.5x	21.0x	4.3x

A relative valuation comparison further supports Teradyne's investment appeal - reflecting a balanced premium to more cyclical peers like AMAT (13.8x EV/EBITDA) and Keysight (16.5x), while remaining at a meaningful discount to high-growth peer Advantest (39.7x). Teradyne's valuation is justified by its strong market positioning in semiconductor testing, high free cash flow conversion, and optionality from industrial automation. While it does not yet command AI-fuelled multiples like Advantest, its diversified exposure and earnings stability position it favourably in the mid-cap test equipment space.

The EV/EBITDA sensitivity analysis underscores the upside potential in Teradyne's valuation. At a base case multiple of 18x, the implied share price is \$89.79, representing a ~21% upside from the current price. Even under more conservative scenarios (15x–16x), the downside remains limited, while modest multiple expansion to 20x–21x could drive the stock above **\$100/share**. This highlights a favourable risk-reward profile supported by strong fundamentals and secular tailwinds in semiconductor testing and industrial automation.

Sales (in USB Billions)

Year	Q1	Q2	Q3	Q4	FY
2021	781.6	1085.7	950.5	885.0	3702.9
y-o-y	10.97%	29.46%	15.99%	16.61%	18.63%
2022	755.4	840.8	827.1	731.8	3155.0
y-o-y	-3.36%	-22.56%	-12.99%	-17.31%	-14.79%
2023	617.5	684.4	703.7	670.6	2676.3
y-o-y	-18.25%	-18.59%	-14.91%	-8.37%	-15.17%
2024	599.8	729.9	737.3	752.9	2819.9
y-o-y	-2.87%	6.64%	4.77%	12.27%	5.36%

Diluted EPS

Year	Q1	Q2	Q3	Q4	FY
2021	1.09	1.76	1.41	1.26	5.53
y-o-y	-15.60%	67.62%	14.10%	25.00%	29.21%
2022	0.92	1.16	1.10	1.04	4.22
y-o-y	-45.65%	-34.09%	-29.09%	-30.77%	-23.69%
2023	0.50	0.73	0.78	0.72	2.73
y-o-y	-20.00%	-37.07%	-21.99%	-17.46%	-35.31%
2024	0.40	1.14	0.89	0.90	3.32
y-o-y	52.50%	56.16%	16.53%	20.00%	21.61%

Risk Factors

1. Exposure to Cyclical Demand:

Teradyne is highly sensitive to macroeconomic cycles and semiconductor industry dynamics. Economic slowdowns, overcapacity, and reduced capital expenditures can materially impact revenue and margins. Short order lead times limit demand visibility, while prolonged uncertainty may delay customer investment decisions and slow growth momentum.

2. Customer Concentration Risk:

A significant portion of Teradyne's revenue is concentrated among a few customers. In 2024, its top five customers accounted for 36% of total revenue, with Samsung alone contributing 12.5%. Concentration creates vulnerability to changes in procurement patterns, inventory adjustments, or strategic shifts by major clients.

3. Competitive Intensity:

Teradyne faces intense global competition from both large, well-capitalized incumbents and emerging players, particularly in Asia. It also contends with in-house test solutions developed by leading semiconductor firms. Competitors may introduce products with comparable or superior performance, eroding Teradyne's market share and pricing power.

4. Foreign Exchange Volatility:

While most of Teradyne's test equipment revenue is in USD, its Robotics segment generates a significant share in foreign currencies. A strengthening U.S. dollar could negatively affect reported revenues and margin expansion in international markets, especially in Europe and Asia.

5. Regulatory & Trade Restrictions:

Teradyne is subject to evolving U.S. export controls and trade regulations, particularly with respect to China—an important market for semiconductor capital equipment. These restrictions could limit customer access, delay shipments, or restrict sales of advanced technologies, especially in sensitive verticals like AI and wireless.

6. Integration & Execution Risk in M&A:

With a history of strategic acquisitions (e.g., Universal Robots), Teradyne may pursue further M&A to expand in automation. However, such initiatives carry integration risk, cultural misalignment, and uncertainty in achieving expected synergies—potentially impacting margins and strategic focus.

Porter's 5 Forces

Threat of New Entrants

Low. Entering ATE or industrial robotics requires significant technical knowledge, manufacturing capability, and trust with OEMs. Teradyne's long-standing ties with top-tier clients are hard to replicate. New players face hurdles in patents, export restrictions (especially with China), and regulatory standards. Entry barriers are high due to cost, expertise, and customer stickiness.

Bargaining Power of Suppliers

Moderate. While general electronics are commoditized, precision parts or AI chips may come from a few specialized vendors. Its global supply chain and negotiation power help it manage supplier dependencies. Some specialized inputs may carry power, but Teradyne's scale limits risk.

Bargaining Power of Buyers

Moderate. A few large chipmakers or auto/electronics OEMs dominate Teradyne's revenue, giving them leverage. While it's hard to switch test platforms quickly, buyers demand performance at competitive pricing. Semiconductor capex cycles directly impact order volumes. Teradyne faces strong buyer power due to industry consolidation and capital intensity.

**Threat of
Substitutes**

Moderate. Customers may develop in-house solutions, though this is often costlier and less efficient. Emerging technologies like AI-driven testing and digital twins are enhancing—but not replacing—the need for physical test equipment. Direct substitutes are rare, but evolving tech could reshape test approaches in the long term

**Bargaining Power
of Buyers**

High. Teradyne competes with Advantest, Cohu, National Instruments (now part of Emerson), and Keysight Technologies. The ATE space is concentrated but fiercely innovative. Continuous innovation is essential to keep up with Moore's Law and customers' shrinking product cycles. Teradyne maintains an edge via product quality, precision, and integration with AI/ML and robotics. Despite strong brand and tech moat, rivalry remains high due to innovation pace and global competition.

Environmental, Social and Social Factors (ESG)

Teradyne holds a solid ESG profile with an overall ESG Score of 4.83, ranking as a "Leading" performer among peers. It excels in Governance, scoring 7.81, reflecting robust corporate oversight. The Environmental score of 4.08 and Social score of 3.29 are both rated above median, though the environmental component saw a slight dip (−0.11). Overall, Teradyne shows strong governance and above-average environmental and social performance relative to industry peers.

Environmental (E) Score: 4.08

Teradyne holds an Environmental score of 4.08 (Bloomberg), placing it above median among peers. The company has committed to Net Zero Targets, but has not yet adopted science-based targets. It reports total greenhouse gas (GHG) emissions of 12.42 Mt CO₂e, with intensity metrics of 35.03 (GHG per Sales) and 12.42 (GHG per EVIC). While the company demonstrates progress in emissions transparency, it currently lacks green, social, or sustainability-linked debt—highlighting an opportunity to strengthen its environmental financing strategy.

Social (S) Score: 3.29

Teradyne's social score is 3.29, also rated above median (Bloomberg). Teradyne is dedicated to fostering a positive workplace and supporting surrounding communities. The company engages with educational institutions to develop future talent, particularly in engineering and technology fields. Through initiatives like the Teradyne Grant Program, it supports students and academic professionals working on innovative technical projects. Teradyne has no fossil fuel exposure and no reported violations of the United Nations Global Compact (UNGC), supporting its positioning as a socially responsible organization.

Governance (G) Score: 7.81

Governance is Teradyne's strongest pillar, with a score of 7.81, categorized as Leading (Bloomberg). Teradyne's board is composed of 100% independent directors, ensuring objective oversight. The average tenure of board members is 4.5 years, promoting a balance between fresh perspectives and experienced leadership. The company is favourably rated by third-party ESG assessors, holding an MSCI ESG rating of 'A', a Sustainalytics risk score of 16.07 (low risk), and an S&P Global ESG rank of 51.00. These highlight Teradyne's commitment to strong governance, transparent leadership, and effective oversight.